

精準醫療產業科技人才培訓-系列二
精準醫療分子檢測產業發展與實驗室管理

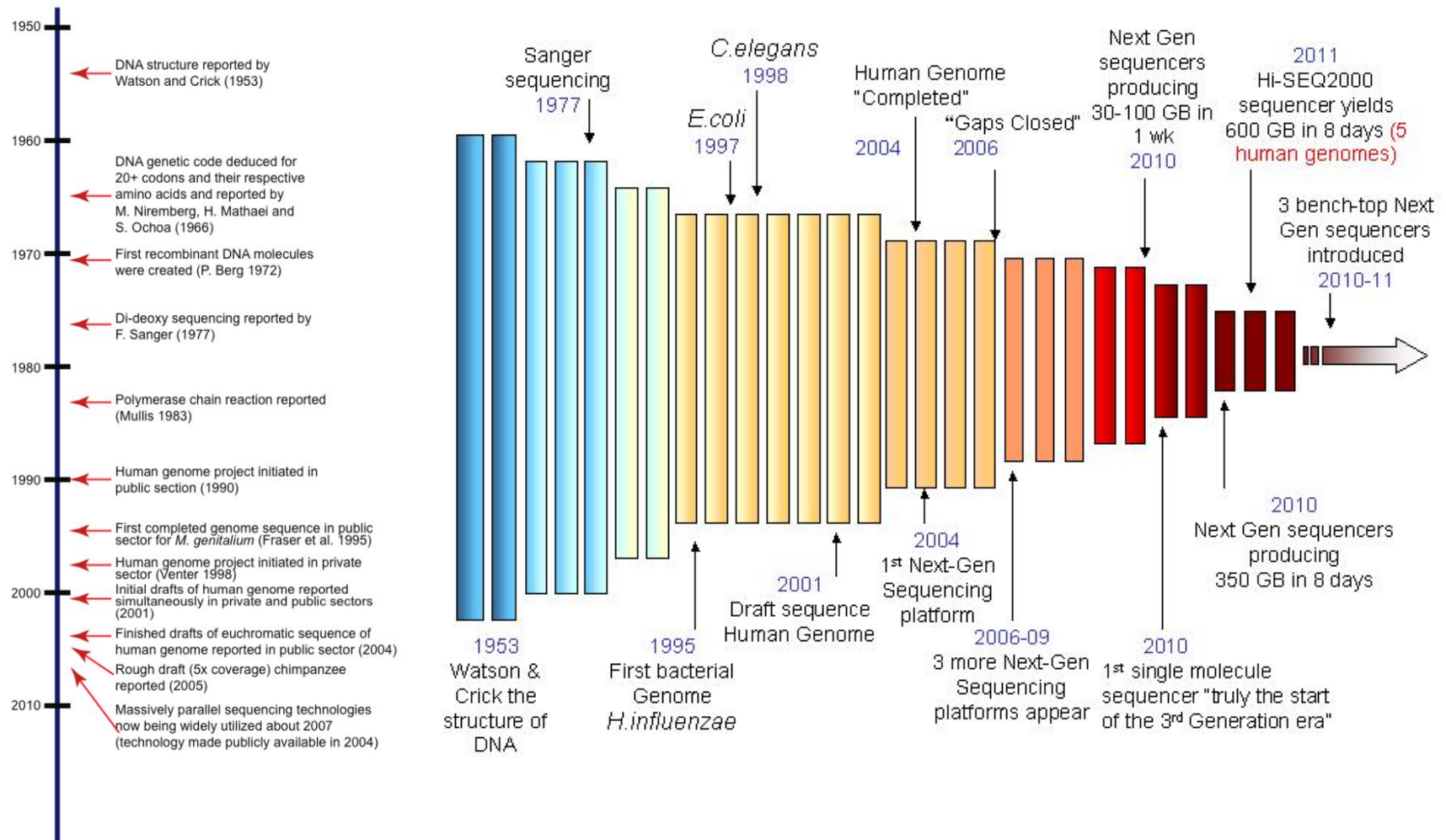
**精準醫療分子檢測之
確效與驗證**

張仕政 博士

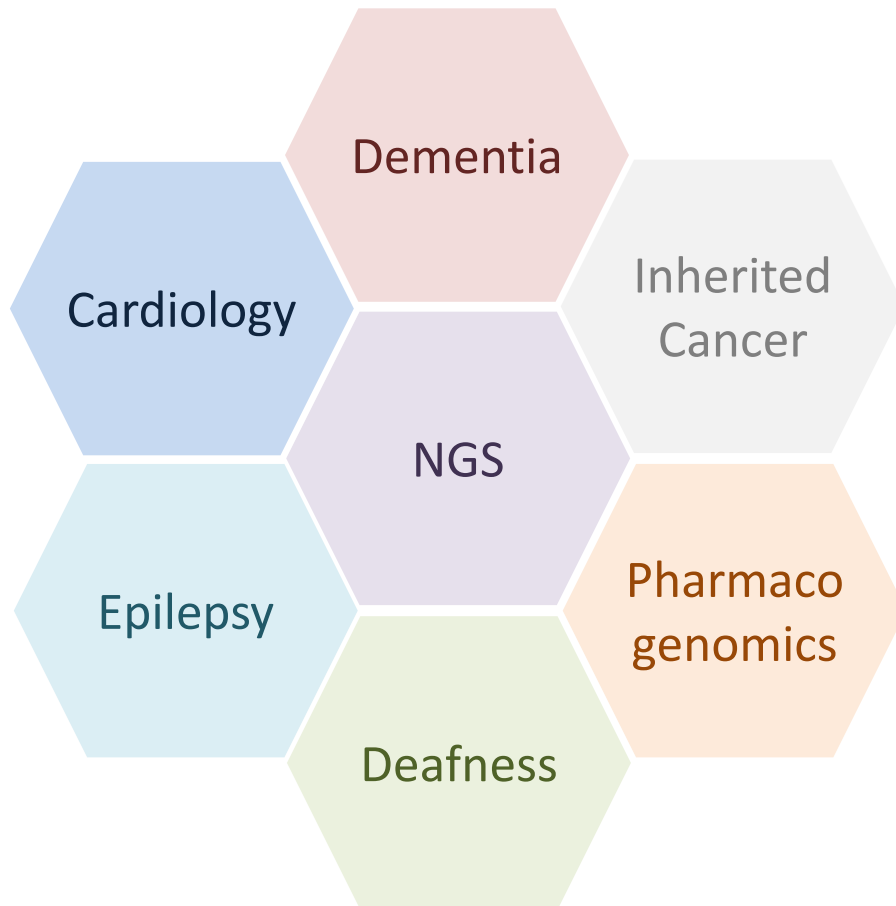
林口長庚醫院 檢驗醫學科 分子檢驗組

2020/07/03

The history of DNA sequencing technology



Germline diseases



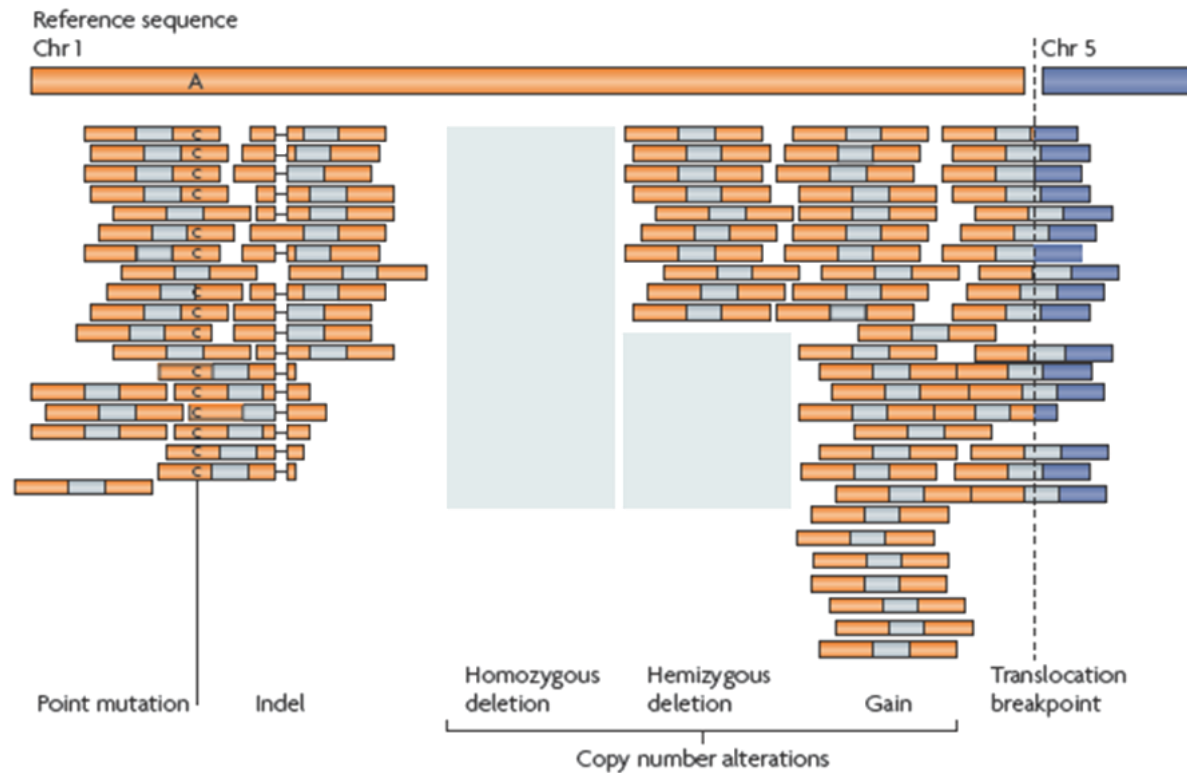
Precision cancer medicine

UNDERSTANDING PRECISION MEDICINE

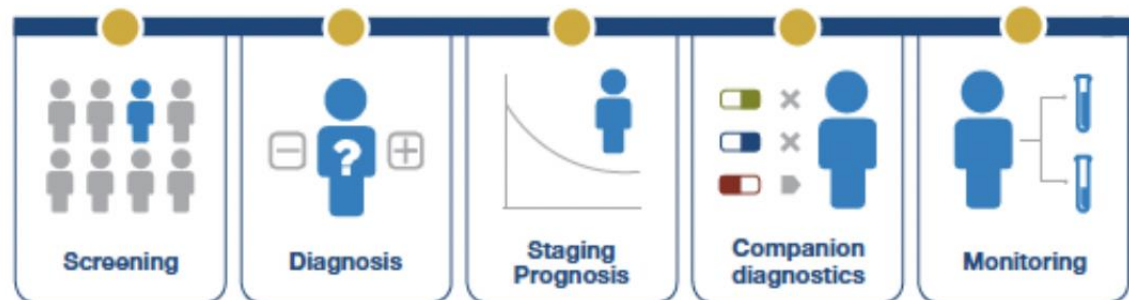
In precision medicine, patients with tumors that share the same genetic change receive the drug that targets that change, no matter the type of cancer.



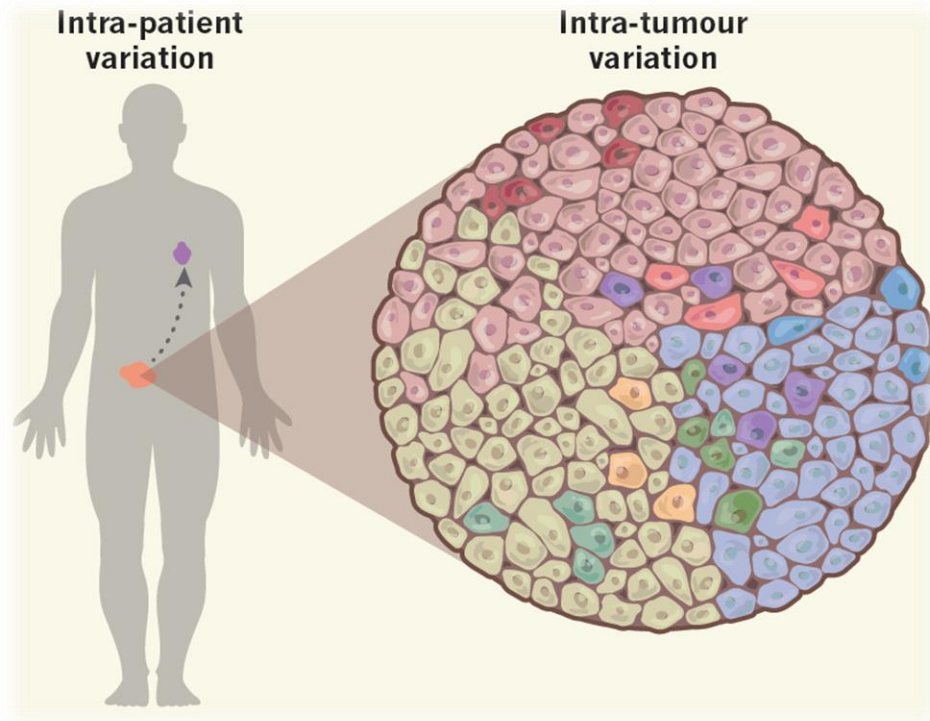
檢驗結果分析技術門檻高



Nature Reviews Genet. 2010 Oct ;11(10):685-96.



同時檢測多種複雜標的

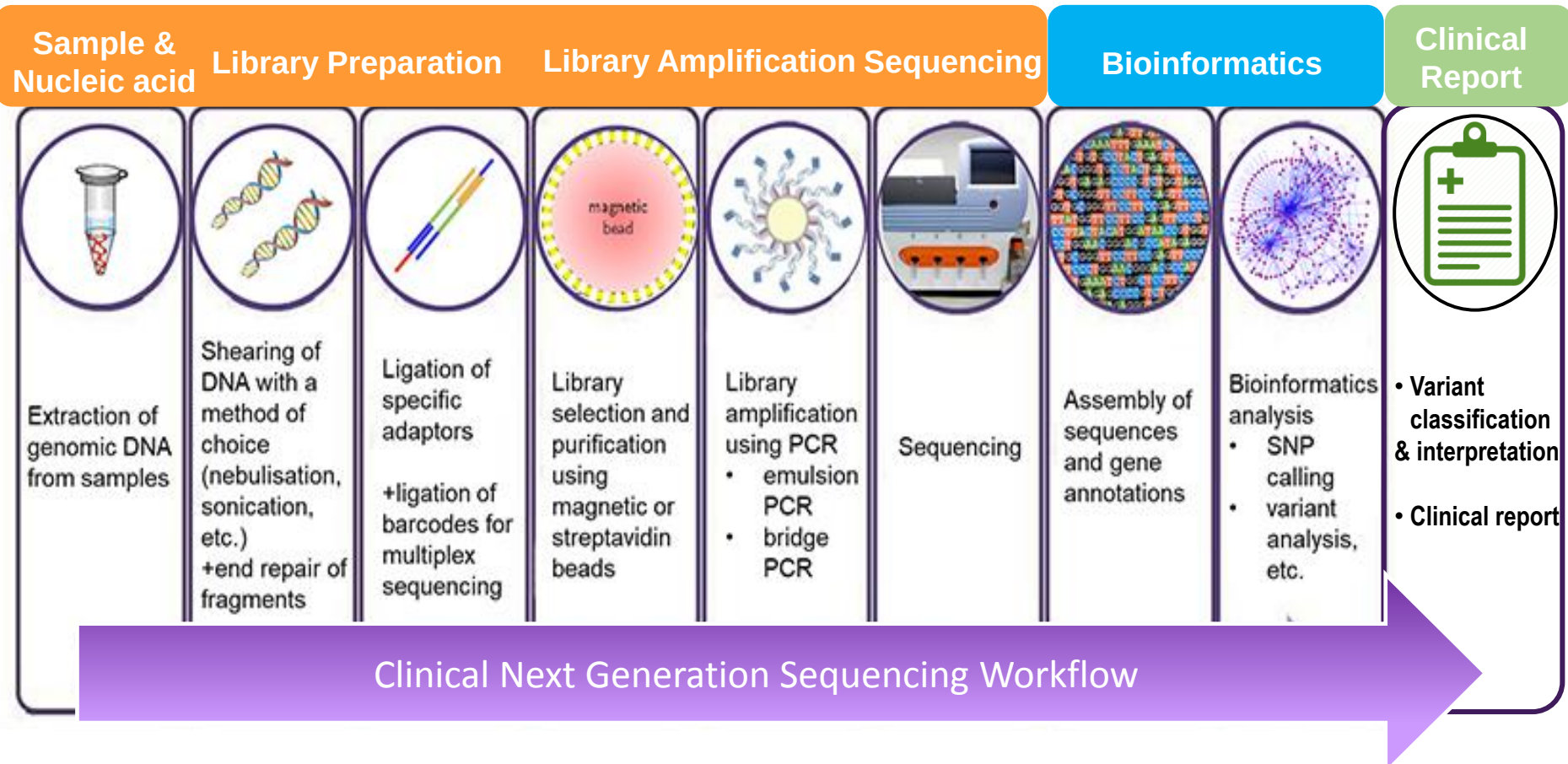


Genetic variations within a tumor:

- Inherited (germline) polymorphisms
- Inherited (germline) variants in cancer-predisposing genes
- Tumor-specific (somatic) variants in oncogenes and tumor suppressors
 - drive the development and growth of the tumor
 - targets of personalized therapies

Adapted from Fox and Loeb, *Nature*, 2014
Jones, *Sci Transl Med.*, 2015

檢驗流程具極高複雜度



Generally, for each test element, test developers should establish metrics and acceptance criteria specific to their particular test as well as the required corrective actions when results fail to meet the laboratory's acceptance criteria

近年歐美地區所公布關於臨床 NGS檢測的指引 (遺傳疾病)

Table 1 Existing Guidelines for NGS Testing (Inherited Disorders)

Year	Organization/entity	PMID	Reference	Title
2010	EuroGentest	20664632	1	A standardized framework for the validation and verification of clinical molecular genetic tests
2012	CDC	23138292	2	Assuring the quality of next-generation sequencing in clinical laboratory practice
2012	AMP	22918138	3	Opportunities and challenges associated with clinical diagnostic genome sequencing: a report of the Association for Molecular Pathology
✓ 2013	ACMG	23887774	4	ACMG clinical laboratory standards for next-generation sequencing
2014	CLSI	NA	5	MM09: nucleic acid sequencing methods in diagnostic laboratory medicine, 2nd edition
✓ 2014	CAP	24650895	6	Methods-based proficiency testing in molecular genetic pathology
2015	CDC	26154004	7	Good laboratory practice for clinical next-generation sequencing informatics pipelines
✓ 2015	CAP	25152313	8	College of American Pathologists' laboratory standards for next-generation sequencing clinical tests
2015	NYS	NA	NA	Guidelines for validation submissions of next generation sequencing (NGS) assays under the NYS testing category of genetic testing— molecular*
✓ 2016	ESHG EuroGentest	26508566	9	Guidelines for diagnostic next-generation sequencing
✓ 2017	CAP	28322587	10	Development and validation of targeted next-generation sequencing panels for detection of germline variants in inherited diseases
2017	CAP	28362156	11	Development and validation of clinical whole-exome and whole-genome sequencing for detection of germline variants in inherited disease
✓ 2018	FDA	NA	NA	Considerations for design, development, and analytical validation of next-generation sequencing (NGS) — based <i>in vitro</i> diagnostics (IVDs) intended to aid in the diagnosis of suspected germline diseases [†]
✓ 2018	AMP/CAP	29154853	12	Standards and guidelines for validating next-generation sequencing bioinformatics pipelines: a joint recommendation of the Association for Molecular Pathology and the College of American Pathologists

遺傳疾病基因變異詮釋

GENETICS in MEDICINE | Volume 17 | Number 5 | May 2015

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ACMG STANDARDS AND GUIDELINES

**Genetics
in Medicine**

Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology

Sue Richards, PhD¹, Nazneen Aziz, PhD^{2,16}, Sherri Bale, PhD³, David Bick, MD⁴, Soma Das, PhD⁵, Julie Gastier-Foster, PhD^{6,7,8}, Wayne W. Grody, MD, PhD^{9,10,11}, Madhuri Hegde, PhD¹², Elaine Lyon, PhD¹³, Elaine Spector, PhD¹⁴, Karl Voelkerding, MD¹³ and Heidi L. Rehm, PhD¹⁵; on behalf of the ACMG Laboratory Quality Assurance Committee

癌症NGS檢測指引

The Journal of Molecular Diagnostics, Vol. 19, No. 3, May 2017



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the Journal of
Molecular
Diagnostics

jmd.amjpathol.org

SPECIAL ARTICLE

Guidelines for Validation of Next-Generation Sequencing—Based Oncology Panels



A Joint Consensus Recommendation of the Association for Molecular Pathology and College of American Pathologists

The Journal of Molecular Diagnostics, Vol. 19, No. 1, January 2017



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the Journal of
Molecular
Diagnostics

jmd.amjpathol.org

SPECIAL ARTICLE

Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer



A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists

NGS Bioinformatics Guidelines

The Journal of Molecular Diagnostics, Vol. 20, No. 1, January 2018



the Journal of
Molecular
Diagnostics

jmd.amjpathol.org

SPECIAL ARTICLE

Standards and Guidelines for Validating Next-Generation Sequencing Bioinformatics Pipelines



A Joint Recommendation of the Association for Molecular Pathology and the College of American Pathologists

Somak Roy,^{*†} Christopher Coldren,^{*‡} Arivarasan Karunamurthy,^{*†} Nefize S. Kip,^{*§¶} Eric W. Klee,^{*||} Stephen E. Lincoln,^{*,**} Annette Leon,^{*††} Mrudula Pullambhatla,^{‡‡} Robyn L. Temple-Smolkin,^{‡‡} Karl V. Voelkerding,^{*§§} Chen Wang,^{*||} and Alexis B. Carter^{*¶¶}

美國食品與藥物管理局2018年指引：

- **Considerations for Design, Development, and Analytical Validation of Next Generation Sequencing (NGS) – Based In Vitro Diagnostics (IVDS) Intended to Aid in the Diagnosis of Suspected Germline Diseases.**
- **Use of Public Human Genetic Variant Databases to Support Clinical Validity for Genetic and Genomic – Based In Vitro Diagnostics.**

衛生福利部食品藥物管理署

精準醫療檢測實驗室檢測技術指引
-次世代定序應用於遺傳性疾病檢測（草案）

目標: 將臨床NGS檢驗標準化

- NGS-based testing represents an evolving technology with continued improvements in instrumentation, sequencing chemistries and bioinformatics and computational analyses
- The guidelines should aim to develop standards that provide a necessary regulatory framework for clinical NGS tests without inhibiting further adoption of NGS-based testing technology
- Fixed thresholds for NGS metrics
- Design concept standards that define approaches and principles to provide guidance

NGS技術應用在臨床檢驗時的 考量要點

- 檢測設計考量 (Test Design Considerations)
- 優化 (Optimization)
- 確效 (Validation)
- 品質管理 (Quality Management)
- 測試性能特徵 (Test Performance Characteristics)
- 實驗品質衡量 (Test Run Quality Metrics)
- 臨床生物資訊 (Clinical Bioinformatics)
- 報告 (Reporting)

檢測設計考量 (Test Design Considerations)

檢測的適應症 (Indication)

- 檢測目的 (應用於疾病之診斷、治療或預後)
- 需有醫學文獻或臨床實證
- 定義檢測基因區段 (全外顯子、目標基因或熱突變點、是否含內顯子、啟動子)
- 定義適用範圍 (適應症、檢測對象、族群分布、特定疾病的流行率等)

檢測設計考量 (Test Design Considerations)

檢體類別 (Specimen type)

- 允收檢體種類 (如血液、口腔拭子、福馬林固定石蠟包埋組織切片等)
- 檢體蒐集容器、檢體穩定性
- 所需最低檢體量
- 單一檢測可能可以檢查不同檢體類別，各類檢體應單獨確效

檢測設計考量 (Test Design Considerations)

次世代定序平台選擇

- 套組大小、基因數目
- 檢測突變形式、通量需求
- 預估檢體量
- 試劑費用與終端價格
- 臨床對於時效之要求
- 資訊分析支援
- 廠商技術更新、機台在使用上的靈活性與擴充性
- 檢驗室本身資源與技術專業程度
- 廠商技術支援

檢測設計考量 (Test Design Considerations)

- 基於可信的科學證據，以及NGS臨床檢測的效力和實用性
- 可影響檢測技術平台的選擇，以及後續確效和品管
- 明確定義適應症和檢測基因標的
- 需記錄於確效評估計畫中

優化 (Optimization)

Optimization and familiarization (O&F)

- 包括NGS所有的流程，從樣品、文庫製備、定序和數據分析
- 在正式的確效開始之前，找出非預期的問題並進行改善
- 提供實驗室技術人員熟悉檢測流程的機會
- 初步性能測試：
 - Well-characterized normal cell lines (the HapMap cell line NA12878, Coriell Cell Repositories)
 - Multiplex DNA sample from Horizon Diagnostics
 - Cell lines with well-characterized alterations
 - Patient specimens of different clinical specimen types
 - Different technologists performing the testing on different days
 - Synthetic BAM or FASTQ files with known mutations to optimize bioinformatics pipeline

確效 (Validation)

- Typically, a new test validation is performed to cover all steps included in the test (DNA extraction → clinical report)
- Validation protocol (確效計畫) should be completed before accumulating validation data
- Data collected during development and optimization are not part of the validation
- Ideally, validation should be performed by the technologists that were not involved in the development or optimization of the test
- Determining types, numbers of samples and the performance specifications

Validation Samples

- Two or more samples which a consensus sequence that has been previously established for all regions covered by the panel
- Previously characterized clinical samples of the specimen type intended for the assay
- Previously characterized clinical samples with each type of pathogenic alteration that the assay is intended to detect (SNVs, indels, CNVs, SVs)
- Samples with most common mutations relevant to the intended clinical use of the panel
- Synthetic BAM or FASTQ files with known SNVs, indels, CNVs, and SVs (for bioinformatics pipeline)
- A minimum of 59 samples to assess quality metrics and performance characteristics (95% confidence, 95% reliability)
- It may be difficult or impossible to find sufficient numbers of samples with larger indels or SVs, the laboratory may choose to supplement the NGS test with another validated test until sufficient number of cases is reached, or include appropriate controls, or clearly state the test limitations within the report.

NGS Reference Materials

Cell Repositories/ Reference Materials

- American Type Culture Collection (ATCC)
- Coriell Cell Repositories
- European Collection of Authenticated Cell Cultures (ECACC)
- Genetic Testing Reference Materials Coordination Program (GeT-RM)
- German Collection of Microorganisms and Cell Cultures
- Japanese Collection of Research Bioresources
- Medical Device Innovation Consortium
- Riken Bioresource Center

Sources of Higher Level Controls (Certified Reference Materials (CRM), Standard Reference materials (SRM))

- Institute for Reference Materials and Measurements
- National Institute of Biological Standards and Controls
- National Institute for Standards and Technology
- National Genetics Reference Laboratory Manchester
- World Health Organization- International Reference Materials

WHO 1st International Standards
for Cancer Genomes

Commercial Sources of Reference Materials (alphabetical order)

This is intended as a list of commercially available stand-alone reference materials available for purchase. AMP does not support or endorse any of the companies or products listed below, but is providing this provider list as a service to the community. If we have omitted your company please [contact us](#).

- Acrometrix
- Advanced Biotechnologies
- Asuragen
- Diagnostic Hybrids
- Horizon Discovery
- Invivoscribe
- LGC Standards
- Maine Molecular Quality Controls
- Molecular Controls
- QIAGEN Marseille (formerly Ipsogen)
- Qnostics
- Seracare Life Sciences
- Vircell Microbiologists
- Zeptometrix

Test Performance Characteristics

準確度 (Accuracy)

- Positive Percent Agreement (PPA)
- Negative Percent Agreement (NPA)
- Technical Positive Predictive Value (TPPV)

分析靈敏度 (Analytical Sensitivity)

- Limit of Detection (LOD)

分析特異性 (Analytical Specificity)

- Interference
- Cross-reactivity
- Cross-contamination

精密度 (Precision)

- Reproducibility
- Repeatability

Clinical Validity (確效性) and Clinical Utility (實用性)

Reportable Range/Reference Range

Validation matrix of NGS

(somatic mutation)

Table 3 Sample Next-Generation Sequencing Validation Matrix

Next-generation sequencing run	Technologist	Sample											
		1	2	3	4	5	6	7	8	9	10	11	12
1	A	PS1	LOD1	PS2	PS3	PS4	LOD1	PS5	PS6	PS7	PS8	PS9	PS10
2	B	PS11	PS12	LOD2	PS13	PS14	PS15	LOD2	PS16	PS17	PS18	PS19	PS20
3	A	PS21	PS22	PS23	LOD3	PS24	PS25	PS26	LOD3	PS27	PS28	PS29	PS30
4	B	PS31	PS32	PS33	PS34	PS35	PS36	PS37	PS38	LOD1	PS39	LOD1	PS40
5	A	PS41	PS42	PS43	PS44	PS45	PS46	PS47	PS48	PS47	LOD2	PS50	LOD2
6	B	LOD3	PS51	PS52	PS53	LOD3	PS54	PS55	PS56	PS57	PS58	PS59	PS60

Technologist A and B represent two individual technologists performing the validation testing. A validation matrix can help ensure that all performance parameters are addressed as efficiently as possible. Testing personnel should be blinded to the identity of samples and controls.

LOD, limit of detection sample; PS, previously tested patient sample.

Each variant type

- **Accuracy** - 60 previous tested patients samples (A minimum of 59 samples)
- **Limit of Detection (LOD)** – 2 of each LOD1-3, and 60 PS
- **Precision** - Technologist A & B (repeatability & reproducibility)
 - LOD1-3 (A minimum of 3 samples)

準確度 (Accuracy)

*Should be calculated for each variant type
(SNV, small indel, larger indel, CNV, structural variants)*

2017 AMP/CAP guidelines for cancer testing

Next-generation sequencing testing result	Orthogonal method positive	Orthogonal method negative	Total
Positive	A	B	A + B
Negative	C	D	C + D
Total	A + C	B + D	A + B + C + D

Positive predictive value
(PPV)=A/(A+B)

Positive percentage agreement (PPA) = $A/(A+C)$; FN=1-PPA

2018 FDA guidance for germline diseases

		Comparator Method		Total
		Positive	Negative	
NGS	Positive	A	B	A+B
	Negative	C	D	C+D
	Total	A+C	B+D	A+B+C+D

Technical PPV= $A/(A+B)$

Negative percent agreement (NPA)= $D/(D+B)$; FP=1-NPA

分析靈敏度 (Analytical Sensitivity)

偵測極限 (*Limit of Detection*)

For germline & somatic mutations

- DNA input: minimum or maximum amount

For somatic mutation (or germline mosaicism)

- Lower LOD: the minor allele fraction at which 95% of samples would reliably be detected
- Establish and document the lower LOD for each variant type included in the test's indications for use
- At least 59 samples should be used
- If sufficient samples cannot be sourced, sensitivity controls (plasmid, other artificial constructs) should be used
- Tumor percentage: Testing specimens with mixed content

分析特異性 (Analytical Specificity)

- 干擾物質 (Interference)

- Endogenous or exogenous substances known to affect molecular testing, particularly amplification-based methods (*heavy metal fixation, melanin, hemoglobin...*)

- 交叉反應 (Cross-Reactivity)

- Repetitive sequence or homologous regions (pseudogenes)

- 交叉汙染 (Cross-Contamination)

- Carryover between patient specimens or samples
- A **no template control (NTC)** should be included in every run during validation and every clinical run (only for amplification steps, need not to be evaluated all the way through sequencing)

精密度 (Precision)

- **重複性 (Repeatability) 確效:** 可使用相同檢體，在不同特定檢驗條件下（如不同操作者、不同操作條件、不同儀器等）進行檢驗，用以測量檢驗的變異性，並說明檢驗變異性的主要來源。
- **再現性 (Reproducibility) 確效:** 為由相同操作者、相同量測系統、相同操作條件、相同地點下，重複測量相同或相似的物體，用以測量檢驗結果的變異性。
- 應計算每個檢體重複檢驗的相同檢驗結果百分比。重複檢驗無召出與無效召出結果的百分比也應報告。
- 應設定重複性與再現性的閾值，並依照每種檢驗條件、受測基因、變異類型分別報告。

A minimum of 3 samples should be tested across all NGS testing steps. A concordance between 95-98% would be satisfactory (Rehm HL, ACMG 2013)

Clinical Validity and Clinical Utility

- 臨床確效性 (Clinical Validity)
 - Clinical Sensitivity
 - Clinical Specificity
- 臨床實用性 (Clinical Utility)
 - Diagnosis
 - Prognostication
 - Making therapeutic decisions
- Need to be defined during design of the test and need to be evaluated during the validation process

Reportable Range and Reference Range

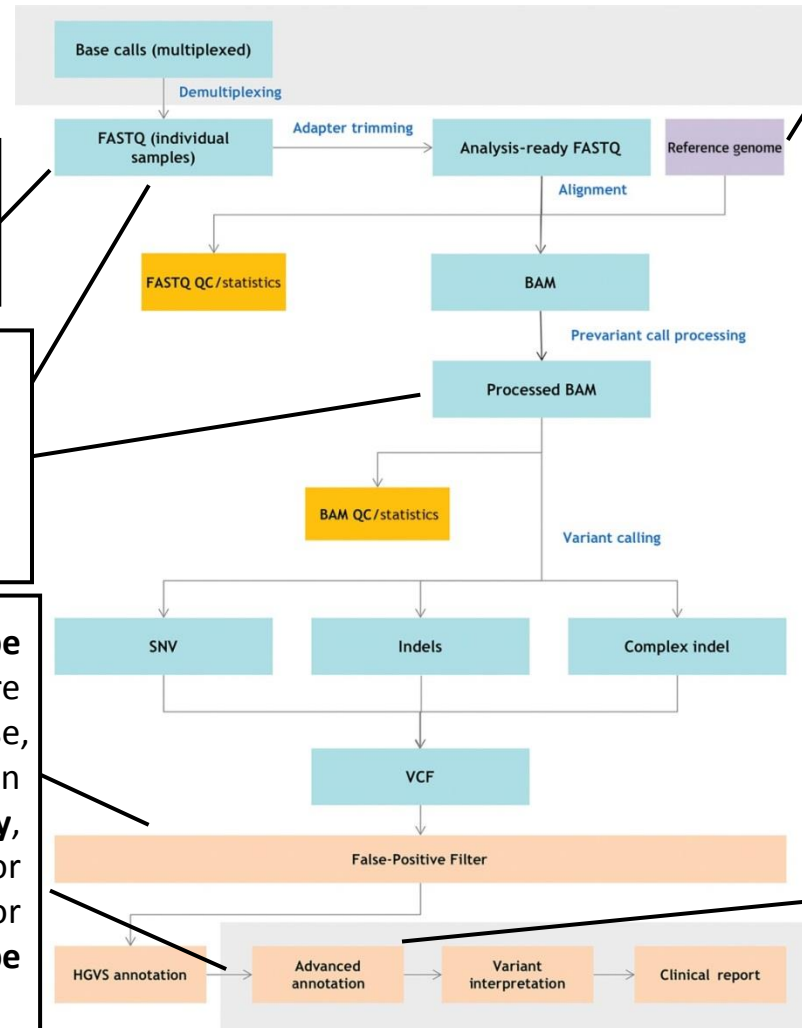
Reportable Range:

- The span of all test results that are considered valid
- Targeted regions that meet the minimum quality requirements
- The variant types that have been validated
- Limits of detection of these variant types

Reference Range:

- For reporting genetic variants, this could be difficult to define
- It should be clearly understood by the ordering provider what types of variants would or would not be reported

NGS Bioinformatics Pipeline



Reference sequence: version number, source...

The laboratory should make a policy regarding the **long-term storage of all relevant data sets**

The **identity** of the sample
 1. A unique sample identifier
 2. A unique patient identifier
 3. A unique run identifier
 4. A laboratory location identifier

The specific **version(s)** of the bioinformatics pipeline used to generate NGS data files are traceable for each patient report (software, database)

The laboratory maintains an **exception log** for patient cases where steps used in the bioinformatics pipeline deviate from standard operating procedures

Structured database (**in-house database**) for all relevant variants with current annotations

How and when filtering will be used. Any filtering criteria that are applied and describe their purpose, e.g. eliminating from consideration variants of **low allele frequency**, **difficult-to-sequence regions** or variants that are **hard to call** or analyze, filtering out **specific type of variant**, etc

Validation should include all individual components of the bioinformatics pipeline used in the analysis (algorithms, software, scripts, reference sequences, databases)

AMP與CAP在生物資訊分析確效的17條建議

Table 2 Consensus Recommendation Statements for NGS Bioinformatics Pipeline Validation

Recommendation number	Statement
1	Clinical laboratories offering NGS-based testing should perform their own validation of the bioinformatics pipeline
2	A qualified medical professional with appropriate training in NGS interpretation and certification must oversee and be involved in the validation process
3	Validation must be performed only after completion of design, development, optimization, and familiarization of the bioinformatics pipeline and its components
4	Bioinformatics pipeline validation should closely emulate the real-world environment of the laboratory in which the test is performed
5	Validation should include all individual components of the bioinformatics pipeline used in the analysis, and each component must be reviewed and approved by an appropriately qualified medical molecular professional and the laboratory director
6	The design and implementation of the bioinformatics pipeline must ensure the security of identifiable patient information and be compliant with all applicable laws at the local, state, and national levels
7	Validation of the NGS bioinformatics pipeline must be appropriate and applicable for the intended clinical use, specimen, and variant types detected of the NGS test
8	Laboratories must ensure that the design, implementation, and validation of the bioinformatics pipeline are compliant with applicable laboratory accreditation standards and regulations
9	The bioinformatics pipeline is part of the test procedure, and its components and processes must be documented according to laboratory accreditation standards and regulations
10	The identity of the sample must be preserved throughout each step of the NGS bioinformatics pipeline with a minimum of four unique identifiers, including a unique location identifier within the content of each data file read and/or generated by the pipeline
11	Specific quality control and quality assurance parameters must be evaluated during validation and used to determine satisfactory performance of the bioinformatics pipeline
12	The methods used to alter or filter sequence reads at any point in the bioinformatics pipeline before interpretation must be validated to ensure that the data presented for interpretation accurately and reproducibly represent the sequence in the specimen, and full documentation of these methods must be kept as part of the test documentation according to laboratory accreditation standards and regulations
13	Laboratories must include specific measures to ensure that each data file generated in the bioinformatics pipeline maintains its integrity and provides alerts for or prevents the use of data files that have been altered in an unauthorized or unintended manner
14	<i>In silico</i> validation can be used to supplement the validation of the bioinformatics pipeline but shall not be used in lieu of end-to-end validation of the bioinformatics pipelines using human samples
15	Validation of the bioinformatics pipeline must include confirmation of a representative set of variants with high-quality independent data; appropriate validation metrics by variant type should be reported
16	Clinical laboratories must ensure the accuracy of software-generated HGVS variant nomenclature and annotations and have an alert in place to indicate when the software-generated nomenclature and annotations need to be manually reviewed and/or corrected, and documentation of any corrections must be maintained
17	Supplemental validation is required whenever a significant change is made to any component of the bioinformatics pipeline

HGVS, Human Genome Variation Society; NGS, next-generation sequencing.

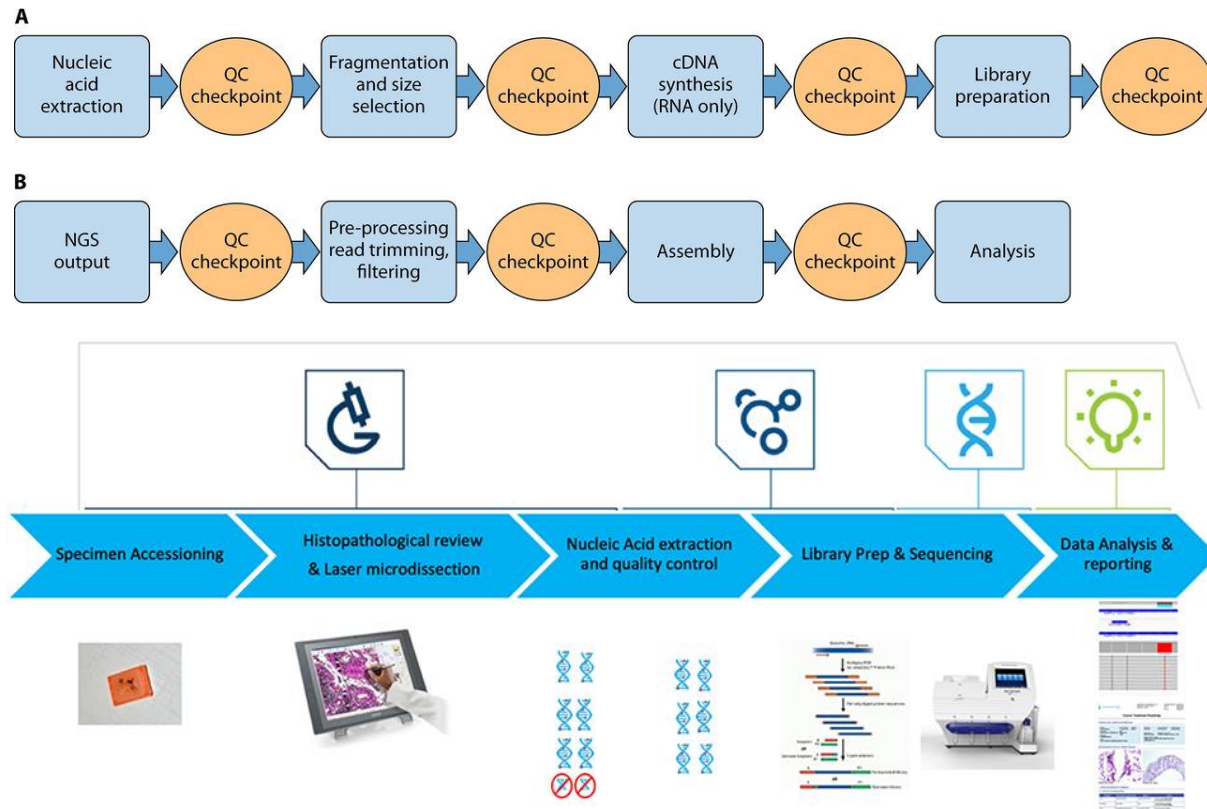
1. 屬於實驗室自身責任
2. 需要**Molecular laboratory director**
3. 在軟體開發測試之後執行
4. 與臨床服務軟硬體條件一致
5. 由**Director**監管所有流程(含外送)
6. 病患資料的使用、傳輸與貯存
7. 預期用途 (**specimen, variant, AF, complex variant**)
8. 需符合實驗室認證與規範
9. 需有**SOP、紀錄與規範**
10. 需有**sample, patient, run, laboratory**辨識碼
11. **QA and QC metrics (Table)**
12. 任何改變或篩選序列片段
13. 確保傳輸完整的資料
14. ***In silico validation***僅為輔助手段
15. 各變異型式的驗證指標
16. **HGVS/Germline or somatic**正確性
17. 分析修改時需補充驗證

Table 4 Recommended Quality Metrics for Clinical Bioinformatics Pipelines

Category	Use	Quality metric	Performance criteria*	Used for
Preanalytical	REQ	% of nucleated cells that are tumor cells	Min	Tumor samples
Sample	REQ	DNA concentration	Min, Max	All sample types
Sample	REQ	DNA fragment size	Min, Max	All sample types
Sample	REQ	Library DNA quantification	Min	All sample types
Run metrics	REQ	Cluster density	Min, Max	All sample types on Illumina platforms that include this metric by default
Run metrics	REQ	% of bases higher than the minimum Phred score of all bases called	Min	All sample types
Run metrics	REQ	Demultiplexing success (ie, all molecular identifiers present and no unexpected molecular identifiers detected)	Pass/fail	All sample types when multiplexing is used
Run metrics	REQ	% of reads passing a minimum Phred score criterion (eg, 99% of bases at Q30 or higher)	Min	All sample types
Read filters	REQ	Mapping quality	Min	All sample types
Mapping [†]	REQ	Mean on-target coverage of reads	Min	All sample types
Mapping [†]	REQ	% of targeted bases with coverage greater than a specified minimum	Min	All sample types
Mapping [†]	REQ	% of bases exceeding the minimum Phred score mapped on target	Min	All sample types
Mapping [†]	OHR	% of aligned bases exceeding the minimum Phred score that disagree with reference	Max	Samples for germline analysis only
Mapping [†]	OHR	AT/GC bias	Max	All sample types
Mapping [†]	REQ	Mean insert size (bp)	Min, Max	All sample types for hybrid capture methods only
Mapping [†]	REQ	% PCR duplicates	Max	All sample types using non-amplicon-based sequencing
Per variant	REQ	Depth of coverage at variant's position	Min	All sample types
Per variant	REQ	Quality score	Min	All sample types
Per variant	Opt	Number of germline SNVs	Min, Max (may have to have separate criteria for different ethnicities)	All sample types
Per variant	REQ	Allele fraction	Min	All sample types
Per variant	REQ	Strand bias	Max	All sample types
Per variant	Opt	Haplotype bias	Max	All sample types
Per variant	REQ	Number of distinct vertical variants at the same position	≤2	All sample types
Per variant	REQ	Number of distinct horizontal variants within a prescribed cluster window size (bp)	≤1	All sample types
QC [†]	OHR	Estimate of % contamination from another sample	Max	Samples for germline analysis only (optional for tumor samples)
QC [†]	Opt	Fingerprint genotypes match NGS results	Yes (no requires investigation/explanation)	All sample types
QC [†]	REQ	Observed sex matches reported sex	Yes (no requires investigation/explanation)	All sample types if X/Y chromosomes are included in assay
QC [†]	Opt	% of bases called that are variants	Min, Max	Samples for germline analysis only (optional for tumor samples)
QC [†]	Opt	SNP/indel ratio	Min, Max	All sample types
QC [†]	Opt	Ti/Tv ratio	Min, Max	All sample types

Establish protocol for entire workflow

- Wet/dry bench SOP
- Daily quality control procedures
- Periodic proficiency testing (PT) or alternate assessment (AA)



後續流程考量建議

重要變異或欲檢驗基因區段的檢驗結果未達預設品質指標時，應建立相關後續流程

- 針對少數未確診疾病，建議執行父母子三人(trio) 試驗或家族試驗，否則，在缺少親子或家族試驗結果時，檢驗結果仍無法判定。
- 在實驗的設計、研發、確效，與上線階段，都需要考量可能的後續流程例如：確認試驗(e.g. Sanger)、補充試驗(fill-in)、父母子三人(trio) 試驗) 。
- 建立後續流程作業規範

ACMG Variant Classification

(Germline disorder)

© American College of Medical Genetics and Genomics **ACMG STANDARDS AND GUIDELINES**

**Genetics
in Medicine**

Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology

Sue Richards, PhD¹, Nazneen Aziz, PhD^{2,16}, Sherri Bale, PhD³, David Bick, MD⁴, Soma Das, PhD⁵, Julie Gastier-Foster, PhD^{6,7,8}, Wayne W. Grody, MD, PhD^{9,10,11}, Madhuri Hegde, PhD¹², Elaine Lyon, PhD¹³, Elaine Spector, PhD¹⁴, Karl Voelkerding, MD¹³ and Heidi L. Rehm, PhD¹⁵; on behalf of the ACMG Laboratory Quality Assurance Committee

Evidence code descriptions of Population data

BA1

Allele frequency is >5% in Exome Sequencing Project, 1000 Genomes Project, or Exome Aggregation Consortium. (Benign, Stand Alone)

BS1

Allele frequency is greater than expected for disorder. (Benign, Strong)

BS2

Observed in a healthy adult individual for a recessive (homozygous), dominant (heterozygous), or X-linked (hemizygous) disorder, with full penetrance expected at an early age. (Benign, Strong)

PM2

Absent from controls (or at extremely low frequency if recessive) in Exome Sequencing Project, 1000 Genomes Project, or Exome Aggregation Consortium. (Pathogenic, Moderate)

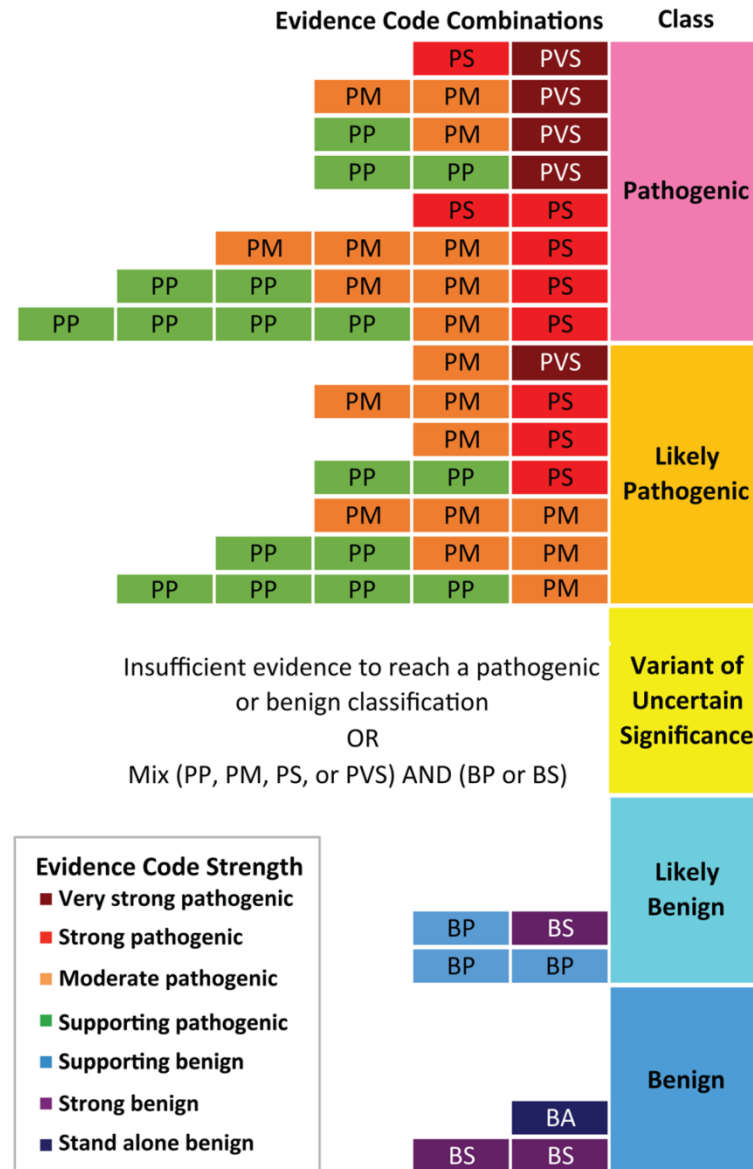
PS4

The prevalence of the variant in affected individuals is significantly increased compared with the prevalence in controls. (Pathogenic, Strong)

Benign		Pathogenic				
	Strong	Supporting	Supporting	Moderate	Strong	Very strong
Population data	MAF is too high for disorder BA1/BS1 OR observation in controls inconsistent with disease penetrance BS2			Absent in population databases PM2	Prevalence in affecteds statistically increased over controls PS4	
Computational and predictive data		Multiple lines of computational evidence suggest no impact on gene /gene product BP4 Missense in gene where only truncating cause disease BP1 Silent variant with non predicted splice impact BP7 In-frame indels in repeat w/out known function BP3	Multiple lines of computational evidence support a deleterious effect on the gene /gene product PP3	Novel missense change at an amino acid residue where a different pathogenic missense change has been seen before PM5 Protein length changing variant PM4	Same amino acid change as an established pathogenic variant PS1	Predicted null variant in a gene where LOF is a known mechanism of disease PVS1
Functional data	Well-established functional studies show no deleterious effect BS3		Missense in gene with low rate of benign missense variants and path. missenses common PP2	Mutational hot spot or well-studied functional domain without benign variation PM1	Well-established functional studies show a deleterious effect PS3	
Segregation data	Nonsegregation with disease BS4		Cosegregation with disease in multiple affected family members PP1	Increased segregation data →		
De novo data				De novo (without paternity & maternity confirmed) PM6	De novo (paternity and maternity confirmed) PS2	
Allelic data		Observed in <i>trans</i> with a dominant variant BP2 Observed in <i>cis</i> with a pathogenic variant BP2		For recessive disorders, detected in <i>trans</i> with a pathogenic variant PM3		
Other database		Reputable source w/out shared data = benign BP6	Reputable source = pathogenic PP5			
Other data		Found in case with an alternate cause BP5	Patient's phenotype or FH highly specific for gene PP4			

Figure 1 Evidence framework. This chart organizes each of the criteria by the type of evidence as well as the strength of the criteria for a benign (left side) or pathogenic (right side) assertion. Evidence code descriptions can be found in [Tables 3](#) and [4](#). BS, benign strong; BP, benign supporting; FH, family history; LOF, loss of function; MAF, minor allele frequency; path., pathogenic; PM, pathogenic moderate; PP, pathogenic supporting; PS, pathogenic strong; PVS, pathogenic very strong.

Rules for combining criteria to classify sequence variants



ACMG Recommendations for Reporting of Incidental Findings in Clinical Exome and Genome Sequencing

(ACMG SF v2.0 - 59 medically actionable genes)

Table 1 ACMG SF v2.0 genes and associated phenotypes recommended for return of secondary findings in clinical sequencing

Phenotype	MIM disorder	PMID Gene Reviews entry	Typical age of onset	Gene	MIM gene	Inheritance ^a	Variants to report ^b
Hereditary breast and ovarian cancer	604370 612555	20301425	Adult	<i>BRCA1</i> <i>BRCA2</i>	113705 600185	AD	KP and EP
Li-Fraumeni syndrome	151623	20301488	Child/adult	<i>TP53</i>	191170	AD	KP and EP
Peutz-Jeghers syndrome	175200	20301443	Child/adult	<i>STK11</i>	602216	AD	KP and EP
Lynch syndrome	120435	20301390	Adult	<i>MLH1</i> <i>MSH2</i> <i>MSH6</i> <i>PMS2</i>	120436 609309 600678 600259	AD	KP and EP
Familial adenomatous polyposis	175100	20301519	Child/adult	<i>APC</i>	611731	AD	KP and EP
<i>MYH</i> -associated polyposis; adenomas, multiple colorectal, <i>FAP</i> type 2; colorectal adenomatous polyposis, autosomal recessive, with pilomatricomas	608456 132600	23035301	Adult	<i>MUTYH</i>	604933	AR ^c	KP and EP
Juvenile polyposis	174900	20301642	Child/adult	<i>BMPR1A</i> <i>SMAD4</i>	601299 600993	AD	KP and EP
Von Hippel-Lindau syndrome	193300	20301636	Child/adult	<i>VHL</i>	608537	AD	KP and EP
Multiple endocrine neoplasia type 1	131100	20301710	Child/adult	<i>MEN1</i>	613733	AD	KP and EP
Multiple endocrine neoplasia type 2	171400 162300	20301434	Child/adult	<i>RET</i>	164761	AD	KP
Familial medullary thyroid cancer ^d	1552401	20301434	Child/adult	<i>RET</i>	164761	AD	KP
<i>PTEN</i> hamartoma tumor syndrome	153480	20301661	Child/adult	<i>PTEN</i>	601728	AD	KP and EP
Retinoblastoma	180200	20301625	Child	<i>RB1</i>	614041	AD	KP and EP
Hereditary paraganglioma-pheochromocytoma syndrome	168000 (PGL1) 601650 (PGL2) 605373 (PGL3) 115310 (PGL4)	20301715	Child/adult	<i>SDHD</i> <i>SDHAF2</i> <i>SDHC</i> <i>SDHB</i>	602690 613019 602413 185470	AD	KP and EP KP KP and EP
Tuberous sclerosis complex	191100 613254	20301399	Child	<i>TSC1</i> <i>TSC2</i>	605284 191092	AD	KP and EP
WT1-related Wilms tumor	194070	20301471	Child	<i>WT1</i>	607102	AD	KP and EP
Neurofibromatosis type 2	101100	20301380	Child/adult	<i>NF2</i>	607379	AD	KP and EP
Ehlers-Danlos syndrome, vascular type	130050	20301667	Child/adult	<i>COL3A1</i>	120180	AD	KP and EP
Marfan syndrome, Loays-Dietz syndromes, and familial thoracic aortic aneurysms and dissections	154700 609192 608967 610168 610380 613795 611788	20301510 20301312 20301299	Child/adult	<i>FBN1</i> <i>TGFBR1</i> <i>TGFBR2</i> <i>SMAD3</i> <i>ACTA2</i> <i>MYH11</i>	134797 190181 190182 603109 102620 160745	AD	KP and EP
Hypertrophic cardiomyopathy, dilated cardiomyopathy	115197 192600 601494 613690 115196 608751 612098 600858 301500 608758 115200	20301725	Child/adult	<i>MYBPC3</i> <i>MYH7</i> <i>TNNI2</i> <i>TNNI3</i> <i>TPM1</i> <i>MYL3</i> <i>ACTC1</i> <i>PRKAG2</i> <i>GLA</i> <i>MYL2</i> <i>LMNA</i>	600958 160760 191045 191044 191010 160790 102540 602743 300644 160781 150330	AD XL AD	KP and EP KP KP and EP KP KP and EP (hemi, het, hom) KP KP and EP
Catecholaminergic polymorphic ventricular tachycardia	604772			<i>RYR2</i>	180902	AD	KP

Sample reports

Gene Transcript	Location	Variant	Zygosity	Classification	Disease	Inheritance	Parental Origin
YY NM_12345.6	Exon X/ Intron X/ Promoter/ UTR	c.XXX (p.XXX)	(Apparently) homozygous/ Heterozygous/ Hemizygous/ Heteroplasmic/ Homoplasmic/ Mosaic/ Somatic	Pathogenic / Likely pathogenic / Uncertain significance	Disease Z	Autosomal recessive/ Autosomal dominant/ X- linked/ Mitochondrial	Paternal/ Maternal/ De novo/ Unknown

Molecular Diagnostic Laboratory

18 Sequencing St. Gene Town, ZY 01234
Tel: 555-920-3333 Fax: 555-920-3334
www.moldlaboratory.com

Patient Name: Jane Doe
DOB: 04/05/1990
Lab Accession: 0123245678
Pedigree #: P9999999
Gender: Female
Race: White

Specimen type: Blood, peripheral
Date specimen obtained: 04/01/2012
Date specimen received: 04/03/2012
Referring physician: John Smith, MD
Referring facility: Regional Hospital
Referring facility MRN: 1225-12251225

TEST PERFORMED - Pan Cardiomyopathy Panel (51 Genes)

INDICATION FOR TEST - Clinical diagnosis and family history of dilated cardiomyopathy (DCM)

RESULT: Positive - An established cause of the reported phenotype was identified

ONA VARIANTS:

RBMT20, Heterozygous c.1913C>T (p.Pro638Leu), Exon 9, Pathogenic
SGCD, Heterozygous c.390delA (p.Ala131fs), Exon 6, Likely Pathogenic
TTN, Heterozygous c.97886G>A (p.Gly32629Asp), Exon 307, Uncertain Significance

INTERPRETATION SUMMARY: This individual carries one previously published pathogenic DCM variant (a missense variant in *RBMT20*). In addition, one loss-of-function variant in the *SGCD* gene was detected. Homozygous loss of function variants in *SGCD* are known to cause Limb-Girdle muscular dystrophy and this individual is likely a carrier for this disease. However, the role of heterozygous *SGCD* variants in autosomal dominant DCM without muscular involvement is not clear and therefore a contribution to disease severity cannot be ruled out. See below for individual variant interpretations.

Cardiomyopathy is typically inherited in an autosomal dominant pattern. Each first-degree relative has a 50% (or 1 in 2) chance of inheriting a variant. *SGCD*-associated Limb-Girdle muscular dystrophy is inherited in an autosomal recessive pattern. Each child of two individuals with pathogenic *SGCD* variants has a 25% (or 1 in 4) chance of having Limb-Girdle muscular dystrophy. Disease penetrance and severity can vary due to modifier genes and/or environmental factors. The significance of a variant should therefore be interpreted in the context of the individual's clinical manifestations.

RECOMMENDATIONS:

Genetic testing of this individual's biological parents and other family members, particularly those who are affected, may help to clarify the significance and relative contributions of the detected variants.

It is recommended that this individual and any 1st degree relative receive continued clinical evaluation and follow-up for features of DCM.

Genetic counseling is recommended for this individual and their family. For assistance in locating nearby genetic counseling services please contact the laboratory at 123-456-7890.

Please note that the classification of variants of unknown significance may change over time if additional information becomes available. Please contact the laboratory at 123-456-7890 once a year for any updates regarding the status of these variants.

INDIVIDUAL VARIANT INTERPRETATIONS:

Pro638Leu in Exon 9 of *RBMT20* (NM_001134363.1) Pathogenic. This variant has been reported in two families with DCM, segregated with disease in >10 affected individuals (including 2 affected obligate carriers), and was absent from 960 race-matched control chromosomes (Brauch 2009). Proline (Pro) at position 638 is highly conserved across evolutionarily distant species and lies within exon 9, which encodes a conserved protein domain where other pathogenic variants in *RBMT20* have been reported (Brauch 2009, Li 2010). In summary, the Pro638Leu variant (*RBMT20*) meets our criteria for pathogenicity based on segregation and absence in controls.

Ala131fs in Exon 6 of *SGCD* (NM_172244.2) Likely Pathogenic. This variant has not been previously reported nor previously identified by our laboratory. This variant is predicted to cause a frameshift, which alters the protein's amino acid sequence beginning at codon 131 and leads to a premature stop codon two amino acids downstream. This alteration is then predicted to lead to a truncated or absent protein (loss of function, LOF). Homozygous LOF variants in the *SGCD* gene have been reported in autosomal recessive Limb-Girdle muscular dystrophy. The clinical significance of a heterozygous LOF variant for DCM in the absence of muscular dystrophy is unknown.

Gly32629Asp in Exon 307 of *TTN* (NM_133378.4) Uncertain Significance. This variant has not been previously reported nor previously identified by our laboratory. Glycine (Gly) at position 32629 is highly conserved in evolutionarily distant species, increasing the likelihood that a change would not be tolerated. Computational tools are mixed on the predicted impact to the protein (AlignGVGD = benign, SIFT = pathogenic), though the accuracy of these tools is unknown. Additional information is needed to assess fully the Gly32629Asp variant.

VARIANTS OF UNLIKELY CLINICAL SIGNIFICANCE:

Variants classified as likely benign or benign are not confirmed by Sanger sequencing. Benign variants are not listed on this report but are available upon request.

The following variants were classified as likely benign because they do not change an amino acid and/or are within a non-conserved region of the intron and/or have a frequency in the general population that strongly argues against a pathogenic effect. While they are more likely benign, the available evidence is insufficient to rule out a disease-causing or contributory role.

DSC2, Heterozygous c.942+12_942+13insTTA, rs35717505, allele freq = none available
MYLK2, Heterozygous c.430C>G (p.Pro144Ala), rs34396614, allele freq = 1.3%, 28/2154 chrom
RBMT20, Heterozygous c.1080A>T (p.Thr360Thr), allele freq = none available
RYR2, Heterozygous c.10254C>T (p.Asn3418Asn), rs138073811, allele freq = none available
TTN, Heterozygous c.18342A>G (p.Lys6114Lys), rs34562585, allele freq = 0.5%, 12/2400 chrom

TEST BACKGROUND: The Pan Cardiomyopathy Panel sequences 46 genes associated with various forms of cardiomyopathy (HCM, DCM, ARVC and LVNC). Cardiomyopathy is typically inherited in an autosomal dominant pattern, though some genes are X-linked. For information regarding the clinical presentation or genetics of a specific type of cardiomyopathy, please visit our website.

TEST METHOD: The coding regions and splice sites of the following 46 genes are completely sequenced in this test: *ABCC9*, *ACTC1*, *ACTN2*, *ANKRD1*, *CASQ2*, *CAV3*, *CRYAB*, *CSRP3*, *CTF1*, *DES*, *DSC2*, *DSP*, *DTNA*, *EMD*, *FHL2*, *GLA*, *JUP*, *LAMA4*, *LAMP2*, *LDB3*, *LMNA*, *MYBPC3*, *MYH6*, *MYH7*, *MYL2*, *MYL3*, *MYLK2*, *MYOZ2*, *NEKN*, *PKP2*, *PLN*, *PRKAG2*, *RBMT20*, *RYR2*, *SGCD*, *TAZ*, *TCAP*, *TMEM43*, *TNNC1*, *TNNI3*, *TNNI2*, *TPM1*, *TTN*, *TTR*, *VCL*. Reference transcripts for each gene can be found on our website. The test is performed by oligonucleotide-based target capture (CaptureReagent, Company) followed by next generation sequencing (Instrument Name). Variant calls are generated using the Burrows-Wheeler Aligner (bwa) followed by GATK analysis. This test detects 100% of substitution variants (95%CI=82-100) and 95% of small insertions and deletions (95%CI=98.5-100). Sanger sequencing is used to provide data for bases with insufficient coverage. All clinically significant and novel variants are confirmed by independent Sanger sequencing. Variants classified as likely benign or benign are not confirmed. This test was developed and its performance characteristics determined by The Molecular Diagnostic Laboratory (CLIA#1111111111). It has not been cleared or approved by the U.S. Food and Drug Administration (FDA).

LIMITATIONS: This test may not detect all variants in non-coding regions that could affect gene expression or copy number changes encompassing all or a large portion of the gene.

REFERENCES:

Brauch KM, Karst ML, Herron KJ, de Andrade M, Pellicka PA, Rodeheffer RJ, Michels VV, Olson TM. 2009. Mutations in ribonucleic acid binding protein gene cause familial dilated cardiomyopathy. J. Am. Coll. Cardiol. 54(10):930-41.

Li D, Morales A, Gonzalez-Quintana J, Norton N, Siegfried JD, Hofmeyer M, Hershberger RE. Identification of novel mutations in *RBMT20* in patients with dilated cardiomyopathy. Clin. Transl. Sci. 2010 Jun;3(3):90-7.

This report was reviewed and approved on May 11th, 2012 3:08 PM by Jane Smith, PhD, FACMG.

Informed Consent and Ethical Considerations

- Limitation
- Variants of unknown significance (VUS)
- Electronic medical records
- Self-provided
- Pretest /post-test counseling
- Medical follow-up
- Incidental findings unrelated to the disease

Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer



A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists

Marilyn M. Li,^{*,†} Michael Datto,^{*,‡} Eric J. Duncavage,^{*,§} Shashikant Kulkarni,^{*,¶} Neal I. Lindeman,^{*,||} Somak Roy,^{*,***}
Apostolia M. Tsimberidou,^{*,††} Cindy L. Vnencak-Jones,^{*,‡‡} Daynna J. Wolff,^{*,§§} Anas Younes,^{*,¶¶} and Marina N. Nikiforova^{*,***}

- An interpretive comment on detected genetic alterations that puts the alteration in **clinicopathologic context** to inform management decisions.
- **HUGO Gene Nomenclature Committee** and **Human Genome Variation Society (HGVS)** variant nomenclature
- **Functional, prognostic, or predictive significance** of the variant for particular tumor type, impact on biochemical pathway(s), and prevalence in relevant cancers.

Evidence-based variant categorization (*Somatic mutations*)

Tier I: Variants of Strong Clinical Significance

Therapeutic, prognostic & diagnostic

Level A Evidence

FDA-approved therapy
Included in professional guidelines

Level B Evidence

Well-powered studies with consensus from experts in the field

Tier II: Variants of Potential Clinical Significance

Therapeutic, prognostic & diagnostic

Level C Evidence

FDA-approved therapies for different tumor types or investigational therapies

Multiple small published studies with some consensus

Level D Evidence

Preclinical trials or a few case reports without consensus

Tier III: Variants of Unknown Clinical Significance

Not observed at a significant allele frequency in the general or specific subpopulation databases, or pan-cancer or tumor-specific variant databases

No convincing published evidence of cancer association

Tier IV: Benign or Likely Benign Variants

Observed at significant allele frequency in the general or specific subpopulation databases

No existing published evidence of cancer association

Reporting of Sequence Variants in Cancer

Tier-based

Relevant Biomarkers

Indicated Contraindicated

Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
KRAS p.(G12S) c.34G>A KRAS proto-oncogene, GTPase Tier: IA Allele Frequency: 3.38%	None	cabozantinib cetuximab ^{1,2} panitumumab ¹ cetuximab + chemotherapy ² panitumumab + chemotherapy ²	37
KIF5B-RET fusion kinesin family member 5B - ret proto-oncogene Tier: IA	cabozantinib vandetanib	None	18

Treatment/
Clinical trials

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
KRAS	p.(G12S)	c.34G>A	COSM517	chr12:25398285	3.38%	NM_033360.3	missense

Gene Fusions (RNA)

Genes	Variant ID	Locus
KIF5B-RET	KIF5B-RET.K15R12.COSF1232	chr10:32317356 - chr10:43612032

Allele freq.
Ref seq.

Relevant Non-Small Cell Lung Cancer Findings

Gene	Finding	Gene	Finding
BRAF	Not detected	NTRK2	Not detected
EGFR	Not detected	NTRK3	Not detected
ERBB2	Not detected	RET	KIF5B-RET fusion
KRAS	KRAS p.(G12S) c.34G>A	ROS1	Not detected
MET	Not detected		

Pertinent
negatives

Clinical values, target region, testing method,
limitation, incidental finding, etc.

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台灣精準醫療品質策進會

Thanks for your attention